KILLING A FINITELY GENERATED IDEAL INSIDE IB BY A SMOOTH MODIFICATION OF THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- One-step lifting through a square-zero ideal. lemma-lift-once (carded L26805). Supplies the factorization C -> B'/J' -> Lambda used inside the construction.
- Equational criterion for flatness. Algebra, lemma flat-eq (cross-chapter, named). Turns the relation sum epsilon_i phi(b_i) = 0 into the displayed system with a_{ij} in R and lambda_j in Lambda.

Source reference. Original reference 00HK: focused-seeds/geometry/S9-smo-lift-twice/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CK: sources/stacks-smoothing/pinned-original-source.tex, source lines 1455–1455. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R \to \Lambda$ be a ring map. Let $I \subset R$ be an ideal. Assume that*

(1) $I^2 = 0$,
(2) $\Lambda/I\Lambda$ *is a filtered colimit of smooth R/I-algebras, and*
(3) $R \to \Lambda$ *is flat.*

Let $\varphi : B \to \Lambda$ be an R-algebra map with B smooth over R. Let $J \subset IB$ be a finitely generated ideal such that $\varphi(J) = 0$. Then there exists R-algebra maps

$$B \xrightarrow{\alpha} B' \xrightarrow{\beta} \Lambda$$

such that B' is smooth over R, such that $\alpha(J) = 0$ and such that $\beta \circ \alpha = \varphi$.

Proof. If we can prove the lemma in case $J = (h)$, then we can prove the lemma by induction on the number of generators of J. Namely, suppose that J can be generated by n elements $h_1, \ldots, h_n$ and the lemma holds for all cases where J is generated by $n-1$ elements. Then we apply the case $n = 1$ to produce $B \to B' \to \Lambda$ where the first map kills of h_n. Then we let J' be the ideal of B' generated by the images of $h_1, \ldots, h_{n-1}$ and we apply the case for $n - 1$ to produce $B' \to B'' \to \Lambda$. It is easy to verify that $B \to B'' \to \Lambda$ does the job.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Assume $J = (h)$ and write $h = \sum \epsilon_i b_i$ for some $\epsilon_i \in I$ and $b_i \in B$. Note that $0 = \varphi(h) = \sum \epsilon_i \varphi(b_i)$. As Λ is flat over R, the equational criterion for flatness (Algebra, Lemma 00HK) implies that we can find $\lambda_j \in \Lambda$, $j = 1, \ldots, m$ and $a_{ij} \in R$ such that $\varphi(b_i) = \sum_j a_{ij}\lambda_j$ and $\sum_i \epsilon_i a_{ij} = 0$. Set

$$C = B[x_1, \ldots, x_m]/(b_i - \sum a_{ij}x_j)$$

with $C \to \Lambda$ given by φ and $x_j \mapsto \lambda_j$. Choose a factorization

$$C \to B'/J' \to \Lambda$$

as in Lemma 07CK. Since B is smooth over R we can lift the map $B \to C \to B'/J'$ to a map $\alpha : B \to B'$. Then $\varphi = \beta \circ \alpha$. To finish the proof we check that $\alpha(h) = 0$. Namely, the fact that α lifts $B \to C \to B'/J'$ implies that

$$\alpha(b_i) = \sum a_{ij}\xi_j + \theta_i$$

for some $\xi_j \in B'$ and $\theta_i \in J' \subset IB'$. Hence we see that

$$\alpha(h) = \alpha(\sum \epsilon_i b_i) = \sum \epsilon_i a_{ij} \xi_j + \sum \epsilon_i \theta_i = 0$$

because of the relations above and the fact that $I^2 = 0$. $\square$